

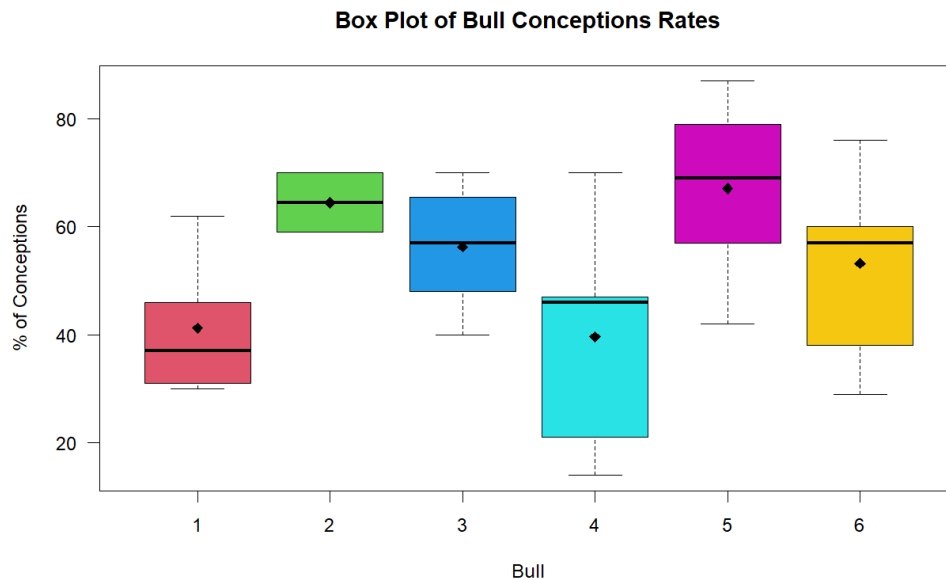
Assignment 1 - 02429 Analysis of correlated data: Linear mixed models

Due date: Monday September 28, 2026

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Exercise 1

Visualisation:



Model

$$y_{ij} = \mu + a_i + \epsilon_{ij}$$

Where:

- y_{ij} represents the j -th percentage measurement for the i -th bull.
- μ is the overall true average percentage of conceptions across the entire population.
- a_i is the random effect for the i -th bull, assumed to be normally distributed with a mean of zero and a variance of σ_a^2 ($a_i \sim N(0, \sigma_a^2)$).
- ϵ_{ij} is the residual measurement error for each specific sample, assumed to be normally distributed with a mean of zero and a variance of σ_ϵ^2 ($\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$).

Results

Intercept (μ) = 53.318

Bull-to-bull variability ($\hat{\sigma}_a^2$): 76.8

Within-bull/Residual variability ($\hat{\sigma}_\epsilon^2$): 248.7

$$h = \frac{4\sigma_a^2}{\sigma_a^2 + \sigma_\epsilon^2}.$$

h = 94.39%

95% Confidence interval of standard deviation of Random Effects

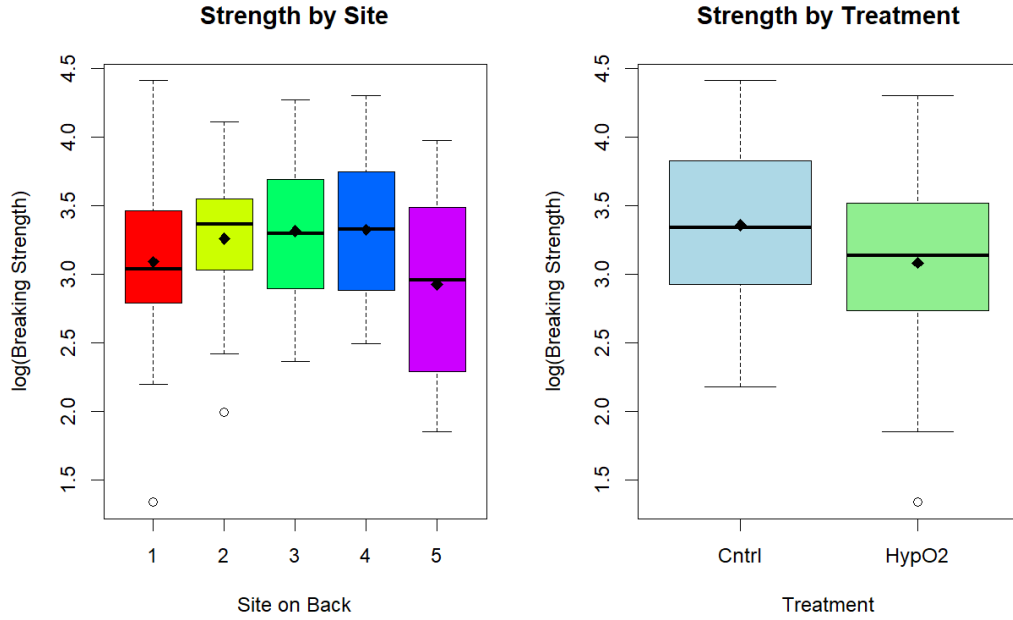
Between Bull-to-bull: (0.000, 19.35)%

Within Bull-to-bull: (12.45, 20.94)%

Overall Average rate of Conception: (43.68, 63.05)%

Summary :

Question 2



Model

Initial Model : With Interactions

$$\log Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + d_k + \epsilon_{ijk}$$

Where:

- $\log Y_{ijk}$ is the log-transformed breaking energy for the k -th rat at the i -th site under the j -th treatment.
- μ is the overall average $\log Y$.
- α_i is the fixed main effect of the i -th site (1 through 5).
- β_j is the fixed main effect of the j -th treatment (Hyperbaric O2 or Control).
- $(\alpha\beta)_{ij}$ is the fixed interaction effect between the i -th site and the j -th treatment, representing how the treatment effect might change depending on the specific site.
- d_k is the random effect of the k -th rat, assumed to be normally distributed with a mean of zero and a variance of σ_{rat}^2 ($d_k \sim N(0, \sigma_{rat}^2)$).
- ϵ_{ijk} is the residual measurement error, assumed to be normally distributed with a mean of zero and a variance of σ^2 ($\epsilon_{ijk} \sim N(0, \sigma^2)$).

Running drop1 shows a p-value value of 0.4292 for the interaction $(\alpha\beta)_{ij}$ so the interaction is insignificant indicating it can be dropped.

Final Model:

$$\log Y_{ijk} = \mu + \alpha_i + \beta_j + d_k + \epsilon_{ijk}$$

```
Random effects:
Groups   Name             Variance Std.Dev.
rat      (Intercept)  0.1476   0.3842
Residual                    0.2110   0.4594
Number of obs: 104, groups: rat, 24

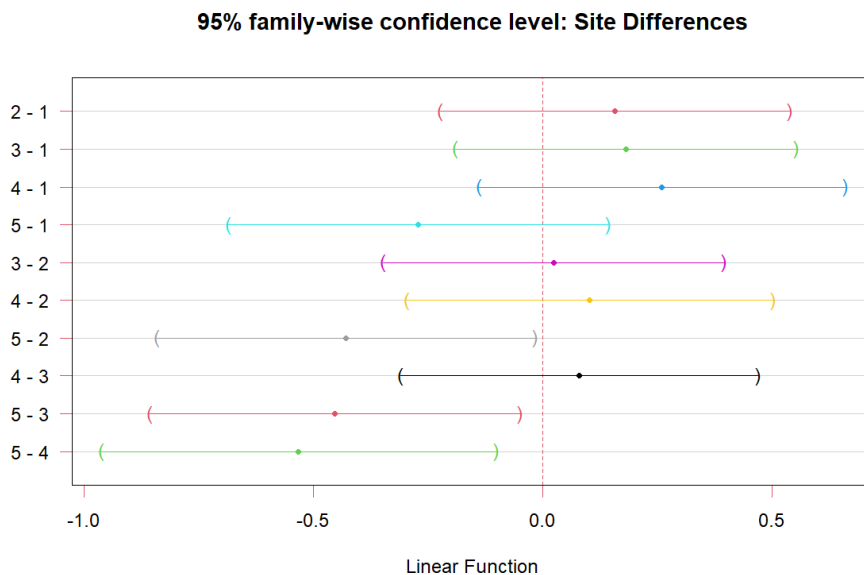
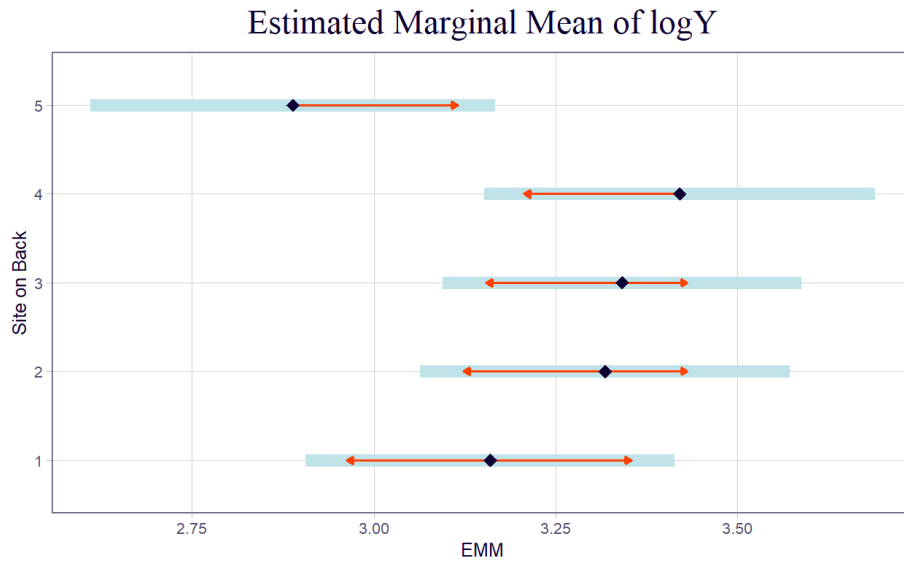
Fixed effects:
              Estimate Std. Error    df t value Pr(>|t|)
(Intercept)   3.3070     0.1659  40.1909  19.934  <2e-16 ***
site2          0.1579     0.1397  76.7186   1.130   0.2621
site3          0.1814     0.1361  76.1183   1.333   0.1866
site4          0.2605     0.1463  77.3744   1.780   0.0790 .
site5         -0.2715     0.1517  78.1823  -1.789   0.0775 .
treatHyp02    -0.2937     0.1843  21.7507  -1.594   0.1254
```

```
Type III Analysis of Variance Table with Satterthwaite's method
              Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
site    3.09214  0.77303     4  77.160   3.6636  0.008736 **
treat    0.53594  0.53594     1  21.751   2.5399  0.125430
```

Part i) As can be seen site has a statistically significant p-value of 0.0087 and is likely to effect the logarithm of the response Y unlike treatment type which has a p-value of 0.1254 which is >0.05 . The variance from rat to rat is 0.1476 and the variance of the residual measurement is 0.2110. The effect of each site can be seen above relative to the intercept which is site 1. Sites 2,3,4 increase the logarithm of the response where site 5 decreases the logarithm of the response. The treatment which is deemed insignificant also decreases the logarithmic the response.

Part ii)

95% Confidence Interval



As can be seen in interactions with site 5, all interactions Confidence intervals don't contain 0, except 5-1 which does contain 0. So there may be no statistical difference between site 5 and 1. But site 5 is different from 2,3 and 4.

Part iii)

Fitting a linear model suggest the treatment is significantly significant due to the linear model underestimating the “true” variance which the LMM takes into account as can be seen below.

State p-values

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.2686    0.1443   22.644  <2e-16 ***
site2         0.1700    0.1784    0.953   0.3431
site3         0.2222    0.1746    1.272   0.2063
site4         0.2615    0.1856    1.409   0.1620
site5        -0.1828    0.1912   -0.956   0.3413
treatHypo2    -0.2977    0.1188   -2.507   0.0138 *
```

Drop1 of linear model:

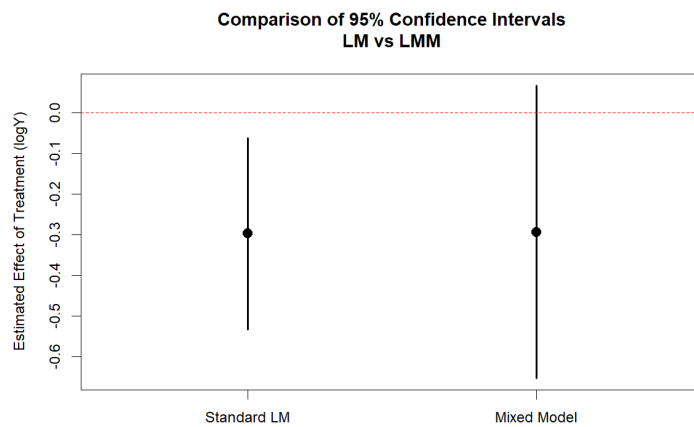
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site    4    2.5265 36.829 -103.962  1.8045 0.13406
treat   1    2.1992 36.502  -98.891  6.2828 0.01384 *
```

Drop1 of LMM:

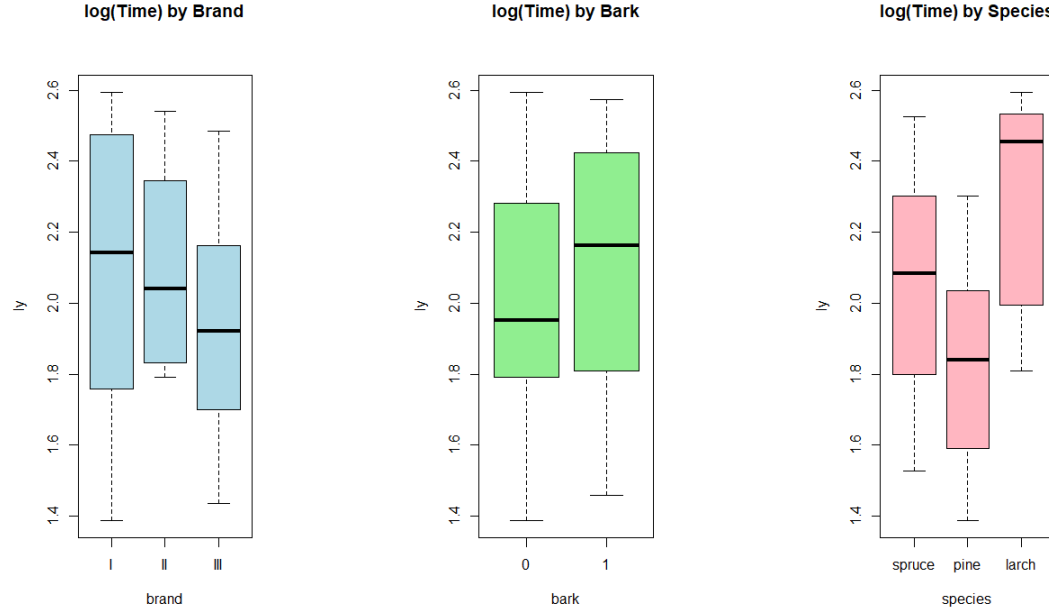
```
> anova(m2)
Type III Analysis of Variance Table with Satterthwaite's method
      Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
site   3.09214  0.77303     4  77.160   3.6636 0.008736 **
treat  0.53594  0.53594     1  21.751   2.5399 0.125430
```

Comparison:

The linear model predicts the treatment is significant with p-value 0.0138 and that site is not with p-value 0.1341. But the LMM predicts the “opposite”. The linear model fits the data



Question 3



Visualisation of Data:

Model:

$$\log(y_{ijklm}) = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \gamma_k + t_l + s_m + \epsilon_{ijklm}$$

Where:

- $\log(y_{ijklm})$ is the log-transformed cutting time for the i -th brand, j -th bark status, k -th species, used by the l -th team, with the m -th specific physical saw.
- μ is the overall average log-transformed cutting time.
- α_i is the fixed main effect of the i -th saw brand.
- β_j is the fixed main effect of the j -th bark status (e.g., bark vs. no bark).
- $(\alpha\beta)_{ij}$ is the fixed interaction effect between the i -th saw brand and the j -th bark status, representing whether the time saved by debarking depends on which brand of saw is used.
- γ_k is the fixed main effect of the k -th wood species (e.g., spruce, pine).
- t_l is the random effect of the l -th team, assumed to be normally distributed with a mean of zero and a variance of σ_{team}^2 ($t_l \sim N(0, \sigma_{team}^2)$).
- s_m is the random effect of the m -th physical saw, assumed to be normally distributed with a mean of zero and a variance of σ_{saw}^2 ($s_m \sim N(0, \sigma_{saw}^2)$).

- ϵ_{ijklm} is the residual measurement error, assumed to be normally distributed with a mean of zero and a variance of σ^2 ($\epsilon_{ijklm} \sim N(0, \sigma^2)$).

Fitting this model shows the interaction is not necessary with a p-value of 0.8730176 on the interaction when performing a F-test

Final Model:

$$\log(y_{ijklm}) = \mu + \alpha_i + \beta_j + \gamma_k + t_l + s_m + \epsilon_{ijklm}$$

Random effects:

Groups	Name	Variance	Std.Dev.
team	(Intercept)	0.079568	0.28208
saw_id	(Intercept)	0.001022	0.03198
Residual		0.008846	0.09405

Number of obs: 36, groups: team, 6; saw_id, 6

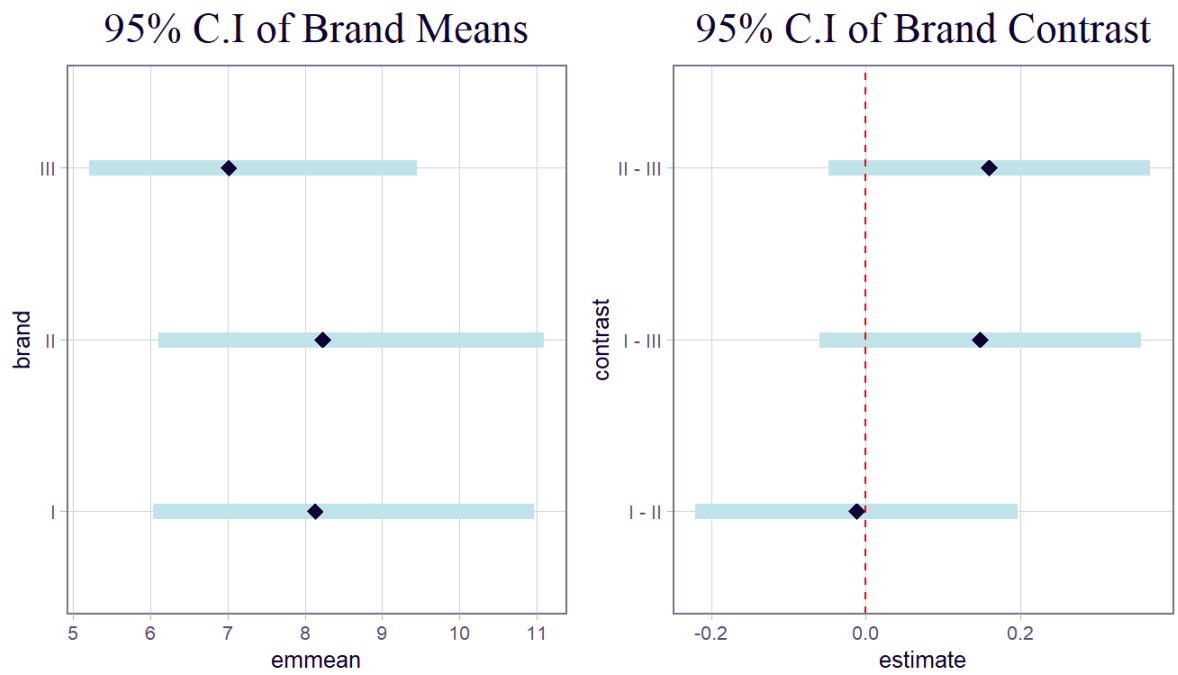
Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	2.00543	0.12348	6.27991	16.241	2.29e-06 ***
brandII	0.01176	0.04997	2.99998	0.235	0.8291
brandIII	-0.14837	0.04997	2.99998	-2.969	0.0591 .
speciespine	-0.21274	0.03840	22.00000	-5.540	1.44e-05 ***
specieslarch	0.25630	0.03840	22.00000	6.675	1.04e-06 ***
bark1	0.15228	0.03135	22.00000	4.857	7.45e-05 ***

Findings:

Fixed Effects : Brand appears to not be significantly significant with the others being significant.

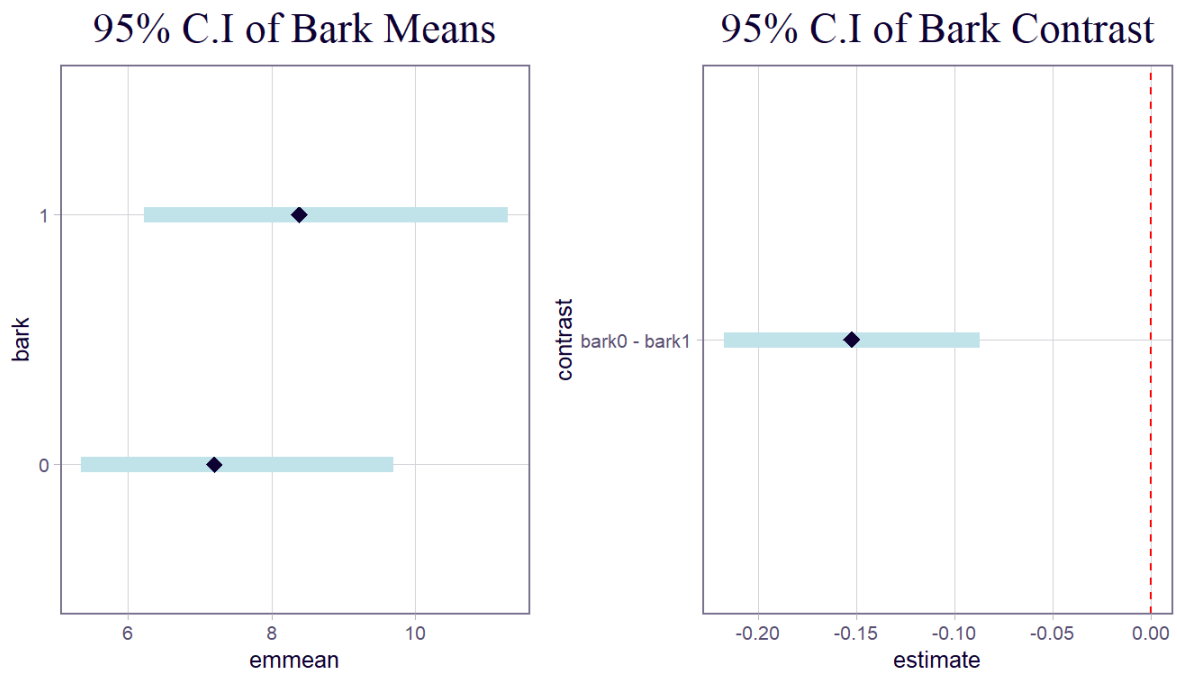
Random Effects : Saw_id has very small variance indicating it has very little effect on the final results. E.g using saw A or D makes very small differences.



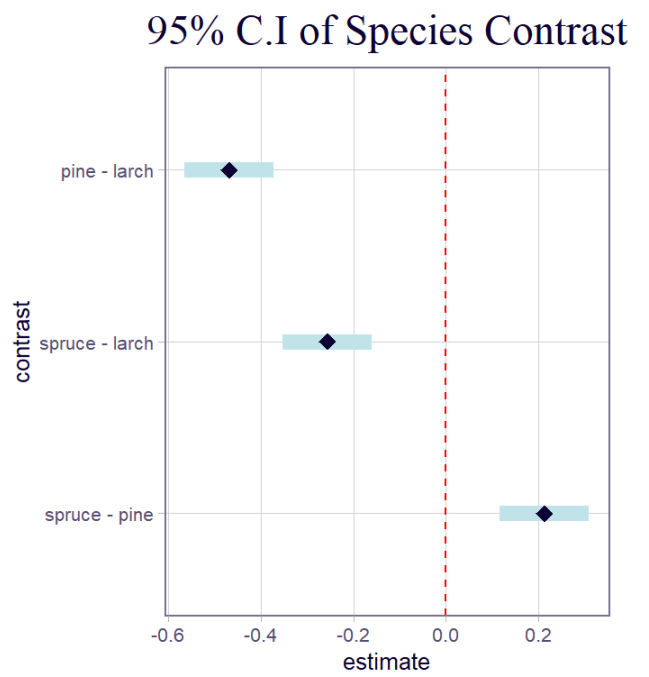
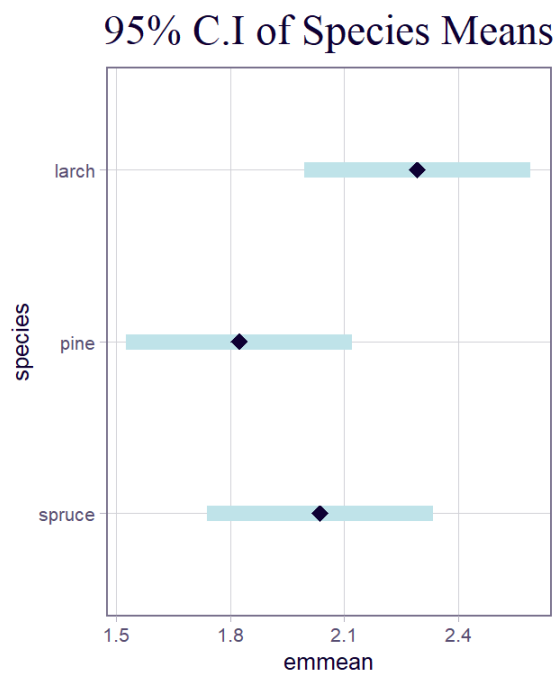
Part i)

As can be seen all brands perform fairly similar and there can be no statistical results in the difference of cutting time due to brand type. Confidence interval of all contrast contains 0.

Part ii)



It can be seen that bark is significantly significant reducing the cutting time by about 0.15 minutes with fairly high confidence not containing 0. Insert p-value



Part iii)

All species are statistically significant. Insert p-values